
Solution to Test

GSM / WCDMA Systems Operation Handling in MSC/BSC

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AXE GENERAL

1. The Application Modules are used to
 - a) **Implement application oriented functions in a flexible way**
 - b) Connect the new hardware.
 - c) Replace the XSS.
 - d) Perform generic (platform) functions
2. OSS and CN-OSS are software packages
 - a) On the AXE for Input Output functions
 - b) On a separate (Unix) platform for AXE management functions
 - c) **On a separate (Unix) platform for Network Management functions**
 - d) On the AXE for Network Management functions
3. AXE MML commands have
 - a) 6 letters and no parameters
 - b) 4 letters and sometimes parameters
 - c) **5 letters and (if present) a parameter block with one or more parameters and (if used) parameter values**
 - d) 5 letters, parameters and parameter values
4. How many Extension Modules can be connected to an RP-pair ?
 - a) 8
 - b) **16**
 - c) 32
 - d) 64
5. Which general types of routes exist in AXE ?
 - a) Outgoing routes, Incoming routes, signaling routes
 - b) Low capacity routes, high capacity routes, very high capacity routes

- c) **External routes, internal routes (announcing machines, signaling) and software routes**
 - d) SS7 routes, CAS routes , traffic routes
6. Name one command to print out the connections of a device
- a) EXDRP
 - b) **EXDEP**
 - c) ANRSP
 - d) ANBSP
7. What are the tasks of the TSMs and the SPMs in the Group Switch ?
- a) **To connect an incoming channel to and an outgoing channel to be able to send speech samples from the A-subscriber to the B-subscriber**
 - b) To switch date and time to the B-subscriber
 - c) To create outgoing and incoming routes routes
 - d) To connect devices for speech transmission
8. Which of both is directly connected to the Group Switch, the SNT or the DIP ?
- a) The Digital Path (DIP)
 - b) **The Switching Network Terminal (SNT)**
 - c) Both
 - d) None of these
9. Which new hardware is implemented in the MSC in the Core Network of WCDMA Systems ?
- a) RPPs in the Packet Control Unit (PCU)
 - b) **ATM Line Interface (ALI), Transcoder and Rate Adapter (TRA) R5B and High Speed Signalling Links (HSL)**
 - c) Echo Canceler Pool (ECP) devices
 - d) CN-OSS

10. Explain the difference between Local and Global Size Alteration Events ?
- a) Local Size Alteration: in one Local Exchange; Global Size Alteration: over more than one exchange
 - b) Local Size Alteration: Size Alteration in one block; Global Size Alteration: over several function blocks;**
 - c) Local Size Alteration: in more function blocks; Global Size Alteration: in one function block
 - d) Local Size Alteration: in more Exchanges in the network; Global Size Alteration: in one exchange
11. What is the function of supervision functions on a route ?
- a) To monitor the traffic load
 - b) To register traffic measurements
 - c) To monitor whether a fault situation occurs**
 - d) To keep data about the traffic for charging purposes
12. What is the normal function of SPG-1?
- a) SPG-1 handles alphanumeric devices and alarms
 - b) SPG-1 is responsible for data communication (DCS)
 - c) SPG-1 is responsible for the connection with the RP Bus
 - d) SPG-1 handles data (charging, statistics)**
13. What is the APG40 used for ?
- a) The Adjunct Processor Group 40 is the replacement of the IOG and is mainly used for IO functions**
 - b) The Adaptive Processor Group 40 is used for Adaptive Multi-Rate (AMR) voice coding
 - c) The Adjunct Processor Group (APG) 40 is connected to the IOG and handles the charging
 - d) The Analog Processing Group is used for subscriber connections
14. What file is always used for an automatic dump from the CP ?
- a) The file with Call Detail Records (CDRs)

b) The file RELFSW0

c) The First File Range RELFSW0 – RELFSW99

d) The Second File Range RELFSW100 – RELFSW127

15. What is the meaning of MTP?

a) Multiple Choice Test Procedure

b) Measurement Timing Program

c) Message Transfer Protocol

d) Measurements Time Protocol

16. On which SS7 layer is the Network Routing handled ?

a) MTP 1

b) MTP 2

c) MTP 3

d) SCCP

MSC BASIC OPERATIONS

17. What is the meaning of the parameter SP in the Routing Case Data ?
- a) Signalling Point
 - b) Switching (Class) Program
 - c) Sending Program**
 - d) Switching Point
18. Which Analysis table is first to be changed in case of implementing new Charging Data ?
- a) Charging Case data table
 - b) Tariff data table**
 - c) Tariff Class data table
 - d) Switching Class data table
19. Which command is used to printout the B-Number Analysis Data ?
- a) ANBSP**
 - b) ANBAP
 - c) ANBPP
 - d) ANBTP
20. In which Analysis table is indicated who is to be charged ?
- a) Charging Case data table**
 - b) Tariff data table
 - c) Tariff Class data table
 - d) Switching Class data table
21. In which Analysis table is decided to route the call to an AST-DR ?
- a) B-Number data table
 - b) End-of-Selection data table**

- c) Routing Case data table
- d) Charging Case data table

22. What is the result of the Pre-Analysis of the B-number ?

- a) Routing Case
- b) Origin for B-Number Analysis**
- c) B-Number Type
- d) Numbering Plan Indicator

23. Which command is used to specify a Traffic Recording Time Schedule ?

- a) TRRPI
- b) TRTSI**
- c) TRRGI
- d) TRTTP

24. What is the purpose of Traffic Measurements ?

- a) To get insight in how the traffic flows
- b) Nice to know
- c) To gather data for Network Capacity Planning and Network Dimensioning**
- d) Just to keep engineers busy.

BSC BASIC OPERATIONS

25. What types of Cell Data do we have in the BSC ?

- a. **Cell Description Data, Cell Definition Data, Location Data**
- b. Cell Data, Managed Objects, Transcoder Devices, Subrate Switch
- c. Tranceiver Group (TG), Receiver (RX), Transmitter (TX), Time Slots (TS)
- d. Cell Global Identity, Cell Size, Transmitting Power,

26. What is a Managed Object in the BSC ?

- a. That refers to Object-Oriented Modelling: an object (unit of software) to be maintained.
- b. **A representation in the BSC of a (part of a) RBS.**
- c. Hardware Unit
- d. Part of the IOG

27. What is representing a Tranceiver Group (TG) ?

- a. A group of RBSs
- b. **One RBS**
- c. Tranceiver Units (TRUs)
- d. A group of carrier frequencies.

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